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Automated Methods for Predicting Wellbore Washouts in Tight Carbonate Rock Based on Drilling Data

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Abstract

The objective of the paper is to develop automated methods for predicting wellbore washouts in tight carbonate formations using high-resolution caliper logs and drilling data. By linking drilling activity patterns with borehole enlargements, the research aims to enhance drilling efficiency and support sustainable drilling practices.

This research utilizes high-resolution caliper logs from Logging-While Drilling (LWD) dual-imaging technology across extended lateral sections (above 10,000 ft) in tight carbonate formation. Data from multiple wells were analyzed, focusing on key drilling parameters like rate of penetration (ROP), and weight-on-bit (WOB). The methodology involves linking high-frequency drilling data to borehole shapes using 180-sector resolution caliper logs. The severity of the impact of different drilling activities on wellbore shapes is estimated, and the risk levels associated with each activity are ranked. Additionally, a predictive model was developed to estimate wellbore size by incorporating activity-based risk levels to closely match the actual wellbore diameter.

The advanced analysis capabilities significantly improved the understanding of drilling impacts on wellbore conditions, ensuring better well placement and minimizing hydrocarbon loss due to wellbore instability. The high-resolution, real-time data provided insights into drilling practices, aiding in future Field Development Planning (FDP) and reducing the environmental footprint associated with drilling risks. Furthermore, the comprehensive view of drilling impacts supported enhanced completion strategies and optimized production, contributing to overall operational efficiency.

The research introduces an innovative approach to predicting wellbore washouts by integrating advanced drilling data with high-resolution caliper logs. This innovative approach is expected to lead to continuous improvement and broader adoption of environmentally responsible drilling practices. By increasing the visibility of environmental benefits and operational efficiencies, this research supports the adoption of advanced digital automation.

Introduction

Drilling operations in tight carbonate formations often face significant challenges, particularly in maintaining a stable and clean wellbore throughout the process. One of the most common issues encountered is wellbore washout, which can lead to difficulties during casing installation, cementing, and even impact the production phase. In certain cases, washouts can already be identified during the drilling phase using caliper data from Logging-While-Drilling (LWD) tools. However, these measurements are typically limited to shortly after the bit has passed and do not fully capture the effects of post-drilling activities, such as reaming, pump cycles, or pulling out of hole. Consequently, further caliper runs are often performed near the end of drilling, using thru-bit or wireline caliper tools, yet these are conducted too late for corrective action for drilling stage, but completion need only.

In this study, we introduce an automated method to predict early signs of wellbore washout by utilizing real-time surface drilling parameter data. This approach was initially explored by correlating the impact of specific drilling activities on borehole caliper readings (*Salahaldeen, 2023*). The study included six wells, each with a total depth exceeding 10,000 feet. These wells were drilled using LWD caliper tools with 180-sector resolution (Figure 1), enabling full circumferential measurement of the borehole geometry. Importantly, caliper measurements were acquired not only during the drilling phase but also during reaming operations and while pulling out of hole, offering a more complete view of borehole conditions. A supporting study in the same field (*Albasser, 2023*) also demonstrated the relationship between borehole stress and changes in borehole size, further reinforcing the methodology presented in this work.

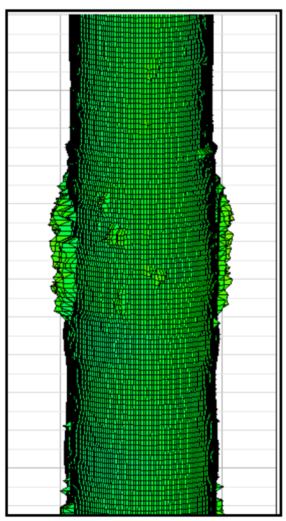

Figure 1—Example of the high-resolution caliper log with a detected wellbore washout

This study incorporates the use of Drill-State classifications, which are commonly available in modern digital drilling platforms. Drill-State refers to specific operational activities, such as rotating, sliding, reaming, circulating, and pulling out of hole (POOH), that provide valuable context to interpret drilling activity. By aligning surface drilling parameters with both time and depth, and mapping them to these refined Drill-State categories, it became possible to identify consistent patterns between certain activity types and the presence of borehole irregularities. These patterns served as the foundation for building an automated model designed to predict zones susceptible to washout in real time. Figure 2 below represents a Drill-State classification in some of drilling monitoring software;

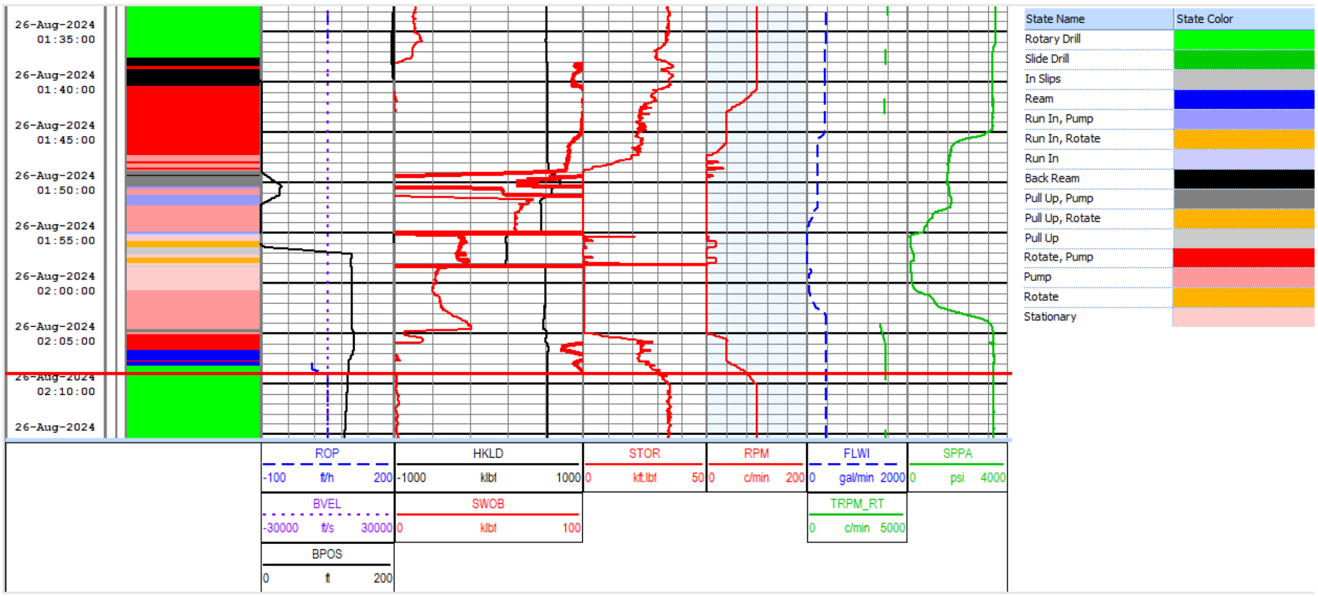

Figure 2—Drill-State timeline from 01:35 to 02:10 on 26-Aug-2024, showing a sequence of eleven distinct drilling activities identified through color-coded classification. The operations include transitions between rotary drilling, back reaming, circulating, running in hole, pulling out, and static condition. This automated rig state provides a clear activity into dynamic downhole behavior and surface response during drilling process.

In addition, a severity matrix was developed to quantify the relative impact of each drilling activity on borehole geometry. The severity matric was created based on the impact of drilling activity to borehole caliper (*Salahaldeen, 2023*). Using the severity scores assigned to each activity, synthetic caliper curves were generated. It is important to note that this method is not designed to replace actual caliper measurements, but rather to complement them, enhancing borehole condition awareness and promoting improved wellbore quality and operational efficiency. This is particularly valuable in tight carbonate formations, where narrow drilling margins and stability challenges require proactive risk management.

Methodology

The methodology is structured into four key stages as described below:

Data Collection from Offset Wells

The initial data collection was evaluated in our previous SPE paper (*Salahaldeen, 2023*), where drilling data were gathered from six offset wells drilled within the same field. All data were captured during active operations with full sensor coverage. Real-time drilling parameters recorded included weight-on-bit (WOB), rate of penetration (ROP), rotary speed, torque, standpipe pressure, and flow rate. These parameters were captured at high frequency and stored in WITSML format. Simultaneously, caliper measurements were acquired using LWD tools equipped with ultrasonic sensors providing 180-sector resolution, enabling detailed borehole shape imaging. Importantly, measurements were obtained not only during drilling but also during reaming and pulling out of hole (POOH), ensuring comprehensive data coverage.

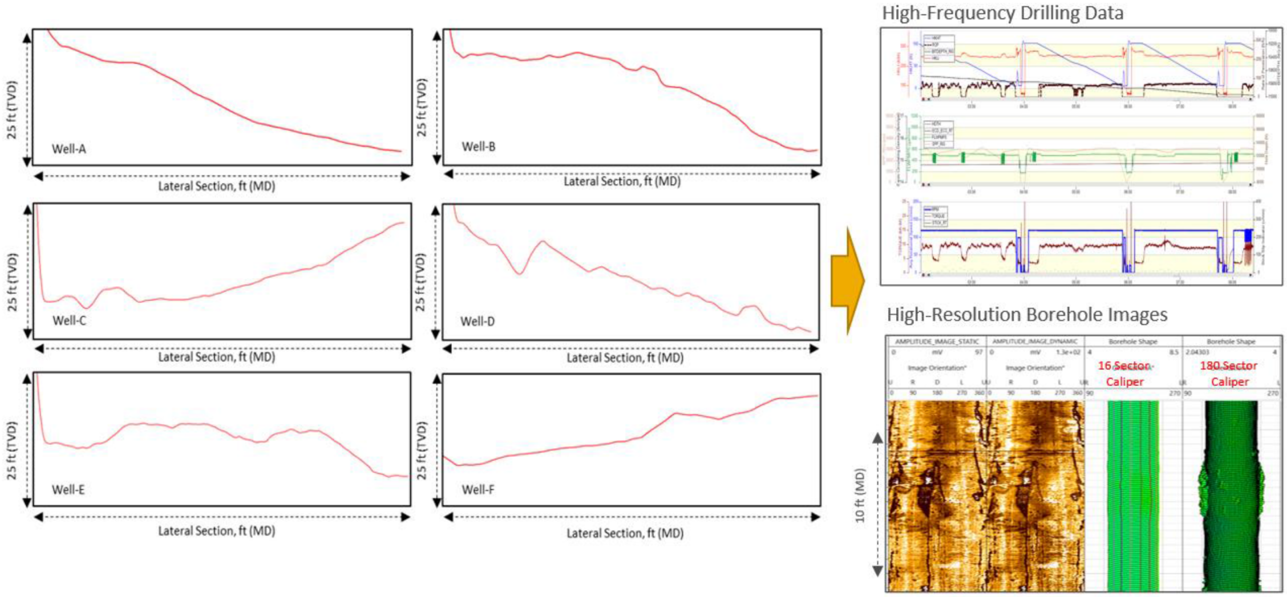

Figure 3—The figure illustrates a typical well used in this evaluation (Salahaldeen, 2023) wellbore quality observations across six wells (Well-A to Well-F). To the right, High-Frequency Drilling Data (e.g., torque, ROP, WOB) is shown, which helps correlate drilling dynamics with downhole conditions. Below this, High-Resolution Borehole Images and caliper data (from 180-sector tools) reveal detailed wellbore shape and breakout patterns. These images highlight severe borehole enlargements and non-circular profiles, which are likely associated with extended reaming/back-reaming.

Data Synchronization and Cleaning

Once collected, surface and downhole data were synchronized by depth and time to ensure accurate correlation between drilling parameters and caliper measurements at each interval. Given the data originated from different systems, adjustments were made to account for delays and sampling frequency variations.

Following synchronization, data cleaning was conducted by removing outliers, such as sudden spikes or drops in WOB or torque that did not reflect actual operations. A moving average or smoothing function was applied to minimize noise while preserving key trends, ensuring data quality suitable for model training.

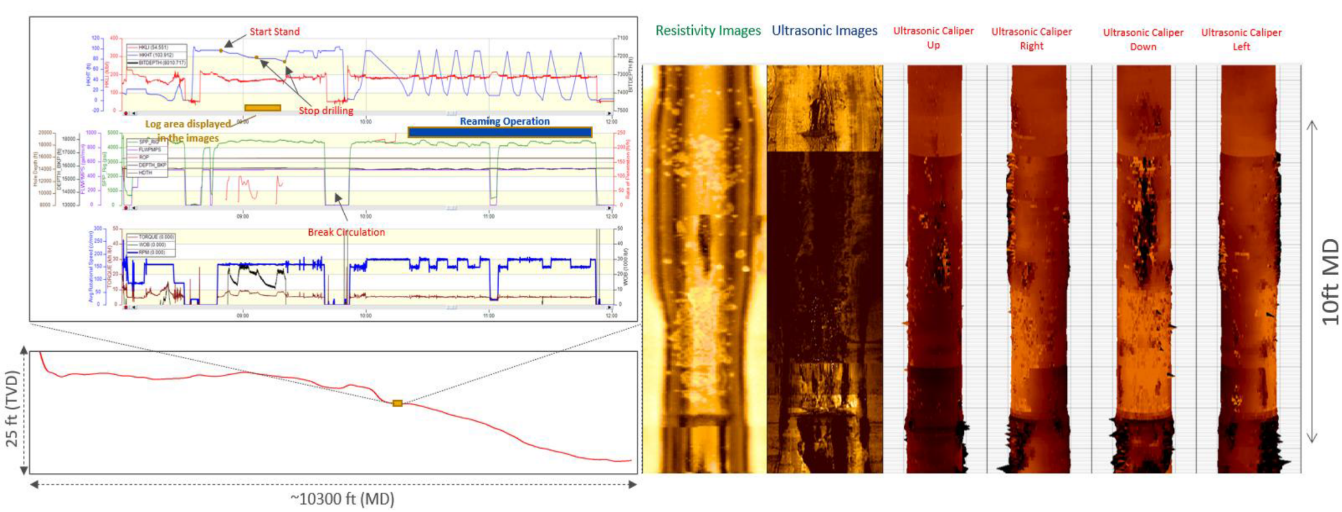

Figure 4—Plot showing before and after data cleaning, for example, ROP and torque signals aligned with caliper diameter across one well section. During drilling at 89° inclination (targeting 90°), operations were halted after advancing 20 ft due to surface equipment issues. To manage the delay, one stand was racked back, and the previous interval was reamed and back reamed for 2 hours. This led to severe wellbore enlargements as seen in the downhole images.

Feature Engineering and Classification

Following synchronization, advanced feature engineering was conducted to develop a robust input set for methods. Nearly 100 unique features were extracted from the high-frequency drilling dataset, including key surface parameters and encoded Drill-State classifications (e.g., rotating, sliding, reaming) as one-hot variables to add operational context. A key addition was the development of a severity-based borehole enlargement impact score, using high-resolution caliper logs. This process involved:

- **Activity-Based Enlargement Classification**: Caliper logs recorded before and after each activity were analyzed to quantify the degree of enlargement.
- **Severity Impact Ranking**: A **severity scale from 1 to 8** was assigned to each type of drilling activity, where a higher value indicates a greater tendency to induce borehole enlargement. This ranking was developed by comparing caliper measurements across intervals subjected to different activities.
- **Formulation for Estimated Diameter**: A predictive formula was built that estimates the borehole diameter based on the assigned activity and its severity rating. The resulting estimated enlargement closely matched the actual caliper readings.
- **Feature Injection**: Integrating the severity score into the feature set to quantify the mechanical impact of drilling activities.

However, in the current field operations, the real-time drilling activity states available (Figure 2) are often generalized and not directly aligned with the classifications we require for accurate borehole washout analysis. Moreover, some of these activity states may not correspond well with the operational behaviors that are critical for borehole condition assessment. Therefore, as part of this study, we introduce a systematic reclassification approach where existing real-time activity states (Figure 2) are reorganized and mapped into a refined activity classification (Figure 5 and Figure 7). The classifications in combined activity (Figure 7) are designed specifically to reflect drilling behaviors that have been previously correlated with washout tendencies, based on historical drilling data and caliper measurements. This reclassification was used to ensure that the observed activities are more relevant to the mechanisms leading to borehole enlargement. Ultimately, this structured activity classification serves as a foundation for integrating into the severity matrix, which can then be used to generate synthetic caliper curves and support proactive washout risk evaluation during drilling operations.

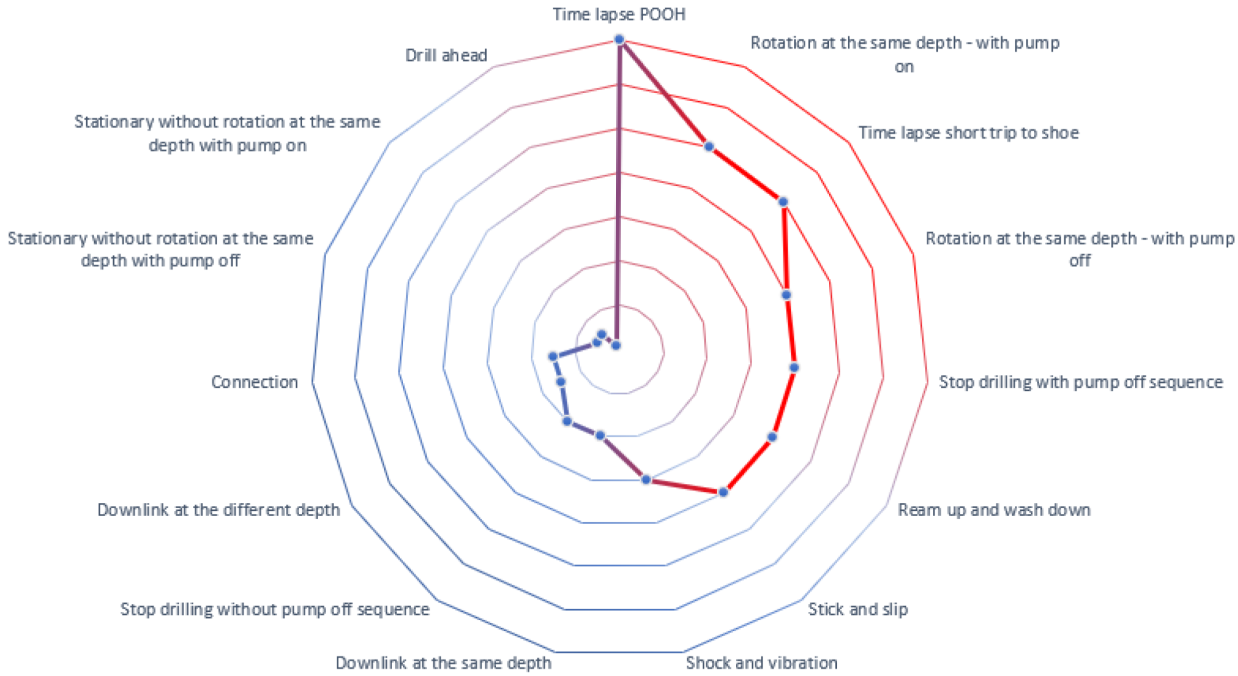

Figure 5—Radar plot illustrating the severity impact scores associated with different drilling activity. The severity scores range from 1 (center) to 8 (outermost ring), with higher scores indicating a greater tendency to induce borehole enlargement. This activity classification was developed by analyzing more than 100 historical drilling samples, combining drilling activity logs and caliper data. The activities were consolidated into 15 operational categories reflecting common drilling behaviors and their respective mechanical impacts on the borehole. This standardized classification provides a critical foundation for linking drilling activities to borehole enlargement tendencies within the severity matrix framework.

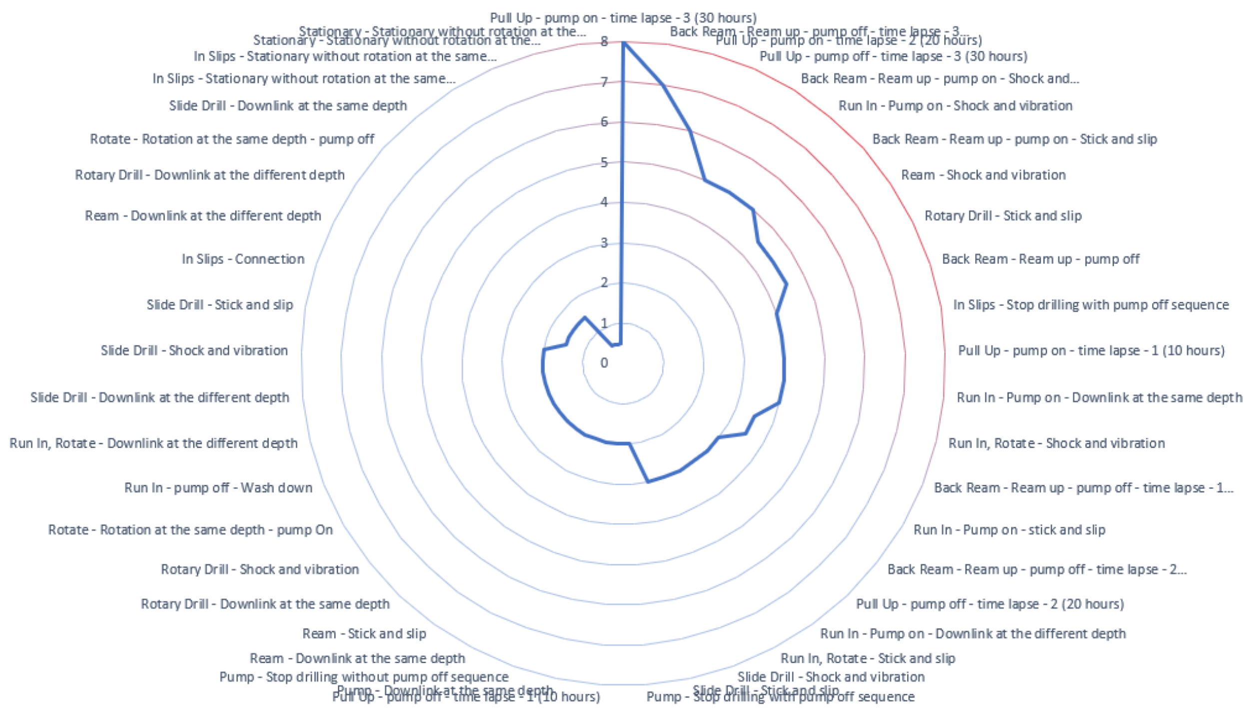

Figure 6—Radar plot showing the severity impact of drilling activities based on combined operational states. The classification was expanded from the initial 15 Drill-State into 45 detailed activity types, reflecting both surface and downhole behaviors. Severity scores (0–8) represent the relative risk of borehole washout, derived from historical drilling and caliper data.

Drill State	Combined Activity
Back Ream	Back Ream - Ream up - pump off
	Back Ream - Ream up - pump off - time lapse - 1 (10 hours)
	Back Ream - Ream up - pump off - time lapse - 2 (20 hours)
	Back Ream - Ream up - pump off - time lapse - 3 (30 hours)
	Back Ream - Ream up - pump on - Shock and Vibration
	Back Ream - Ream up - pump on - Stick and slip
In Slips	In Slips - Connection
	In Slips - Stationary without rotation at the same depth with pump off
	In Slips - Stationary without rotation at the same depth with pump on
	In Slips - Stop drilling with pump off sequence
Pull Up	Pull Up - pump off - time lapse - 1 (10 hours)
	Pull Up - pump off - time lapse - 2 (20 hours)
	Pull Up - pump off - time lapse - 3 (30 hours)
	Pull Up - pump on - time lapse - 1 (10 hours)
	Pull Up - pump on - time lapse - 2 (20 hours)
	Pull Up - pump on - time lapse - 3 (30 hours)
Pump	Pump - Downlink at the same depth
	Pump - Stop drilling with pump off sequence
	Pump - Stop drilling without pump off sequence
Ream	Ream - Downlink at the different depth
	Ream - Downlink at the same depth
	Ream - Shock and vibration
	Ream - Stick and slip
Rotary Drill	Rotary Drill - Downlink at the different depth
	Rotary Drill - Downlink at the same depth
	Rotary Drill - Shock and vibration
	Rotary Drill - Stick and slip
Rotate	Rotate - Rotation at the same depth - pump off
	Rotate - Rotation at the same depth - pump On
Run In	Run In - pump off - Wash down
	Run In - Pump on - Downlink at the different depth
	Run In - Pump on - Downlink at the same depth
	Run In - Pump on - Shock and vibration
	Run In - Pump on - stick and slip
	Run In, Rotate - Downlink at the different depth
	Run In, Rotate - Shock and vibration
	Run In, Rotate - Stick and slip
Slide Drill	Slide Drill - Downlink at the different depth
	Slide Drill - Downlink at the same depth
	Slide Drill - Shock and vibration
	Slide Drill - Shock and vibration
	Slide Drill - Stick and slip
	Slide Drill - Stick and slip
Stationary	Stationary - Stationary without rotation at the same depth with pump off
	Stationary - Stationary without rotation at the same depth with pump on

Figure 7—This figure breaks down the original Drill-State into more detailed combined activities to better capture drilling behavior. Each state, like Back Ream, Pull Up, or Run In, are refined using factors such as pump status, depth changes, shock and vibration, stick and slip, and time lapse. For example, "Back Ream" becomes specific actions like "Ream up – pump off" or "Ream up – pump on – Shock and Vibration." This detailed classification improves accuracy in activity tracking and supports better analysis and automation.

Time Depth Conversion

An important step in the data integration process involved the conversion of Drill-State classifications from the time domain into the depth domain. Drill-State, which represents the sequence of operational activities (e.g., rotating, reaming, pulling out of hole), is typically recorded in time format within most drilling software platforms. However, for the purpose of this study, borehole conditions, including caliper or synthetic caliper data, are evaluated in the depth domain. It was necessary to accurately map the Drill-State data to corresponding depth intervals.

This conversion process was essential to enable the correct association of each drilling activity with the wellbore conditions at specific depths. The Time-Depth Conversion was performed using specialized software tools available from various vendors, which allowed the time-based Drill-State records to be synchronized and converted into depth-based events. This step was particularly challenging in intervals with complex operations, such as extended reaming, back reaming, or static pumping, where the time-depth relationship may become non-linear. Careful validation and mapping adjustments were applied to ensure that all Drill-State classifications accurately reflected the downhole conditions at each depth. This step ensured that the subsequent features of engineering and severity impact assessment were based on correctly aligned and contextually relevant data.

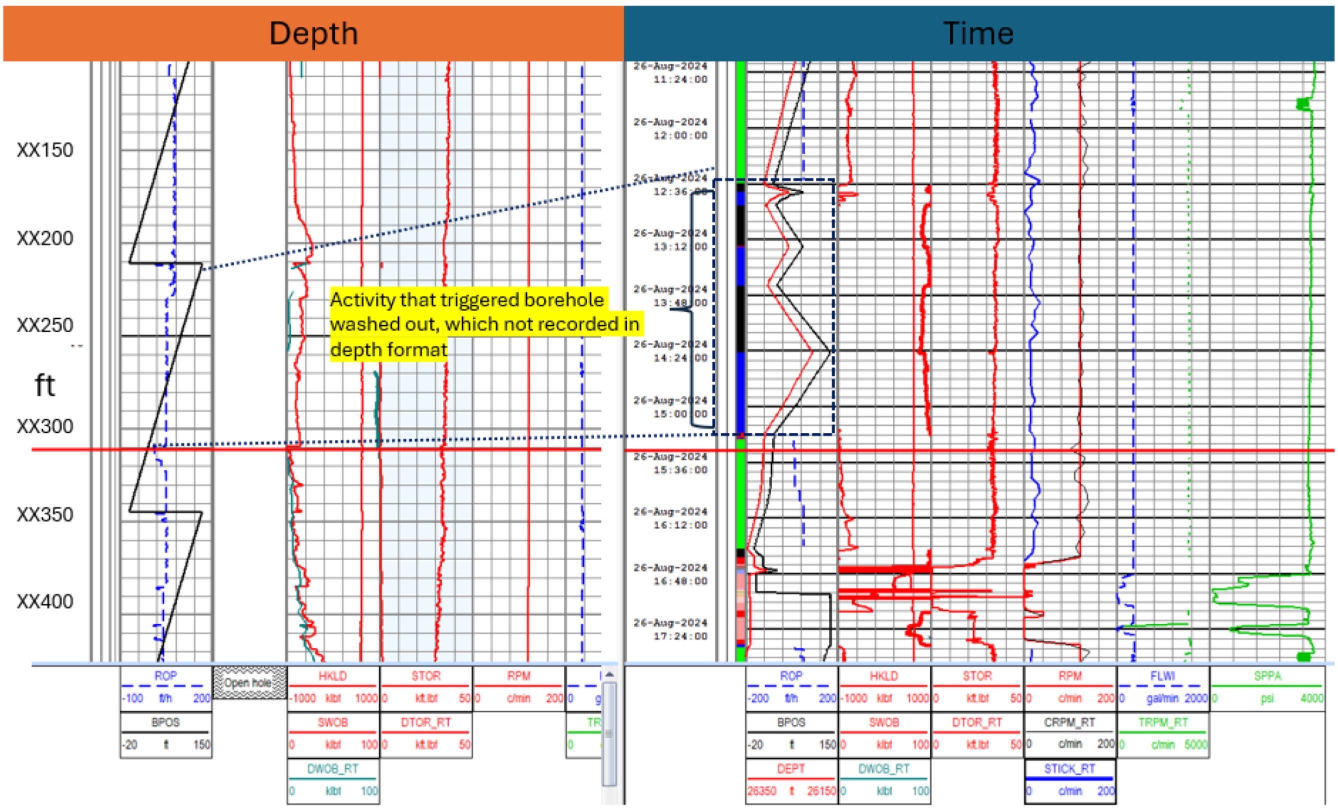

Figure 8—Ream and ream down activity leading to borehole washout. Drilling was paused, halting depth-based data recording. However, Ream up and ream down operations continued at varying depths, captured only in time format (13:36–15:20, 26-Aug-2024). These activities likely triggered borehole enlargement around XX250–XX300 ft. This highlights the importance of converting and integrating time-based events into depth logs for accurate borehole condition analysis.

Results and Discussion

In this initial implementation, the reclassification of the available real-time Drill-State was conducted by manually simulating the refined activity classifications (45 types). This simulation allowed the transformation of the generalized Drill-State into the detailed activity types established in the severity matrix. Based on this reclassified activity, a synthetic caliper log was generated by applying the corresponding severity impact scores. The synthetic caliper curve was then visually compared against the actual caliper measurements recorded by LWD tools during previous drilling operations.

This Figure 9 displays depth and time logs for two distinct wellbore activities, Drilling Pass and the Pull Out of Hole Pass. The yellow-highlighted zones mark depth intervals where the wellbore was exposed during both operational phases (*Noted - Time Log in drilling pass did not cover the whole activity of 300ft due to scale limitation*). The Drill-State color covering in this figure followed the color coding from previous label (Figure 2). These intervals provide insight into how drilling dynamics and borehole conditions may evolve under different mechanical and hydraulic stresses. Notably, variations in parameters such as RPM, WOB, SWOB, torque, and flow rate can be observed when comparing the two passes within these overlapping zones. The alignment of depth and time views enables identification of potential borehole enlargement, tool response consistency, or formation-related events across operations.

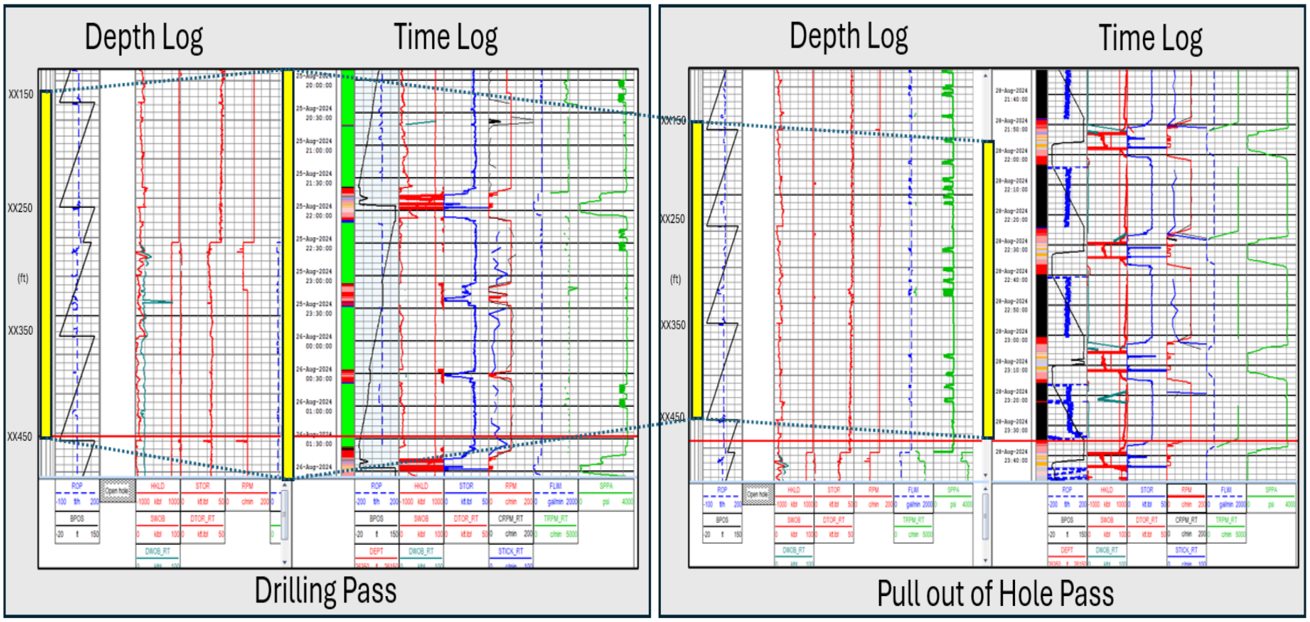

Figure 9—Comparison of Drilling and Pull Out of Hole Passes with Wellbore Exposure Zones

With detail analysist using the updated 45 activity code, its then aligned into both activity. Figure 10 illustrates a comparison between three caliper curves across a lateral interval in a tight carbonate formation. The first column shows manually reclassified Drill-State activity types. These classifications were then used to generate the "Synthetic Caliper Drilling" curve, which estimates borehole diameter based on the severity score associated with each activity type. This visualization supports the methodology's potential for future integration into real-time drilling advisory systems.

Although the process was conducted outside of a real-time drilling software environment, this exercise provided valuable insights into the feasibility of using activity-based classification to predict borehole enlargement tendencies. The comparison between the simulated synthetic caliper and the measured caliper showed encouraging alignment, particularly in zones where washout occurrences. It is important to highlight that this validation remains at a desktop simulation level. The absence of an industry-ready Drill-State refinement system and the limited capability to integrate such workflows directly into drilling operations software are recognized limitations at this stage. Nonetheless, the approach demonstrated the potential of using enhanced activity classification as a basis for wellbore washout risk prediction. Future work will focus on developing automated reclassified Drill-State code and conducting field trials to enable real-time application of this methodology.

Figure 10—Comparison of drilling synthetic caliper, LWD RT caliper, and reaming synthetic caliper. The reprocessed caliper showing larger hole sizes due to post-drilling activities like wash pipe and reaming captured through time-depth conversion.

Conclusions and Recommendations

This study presents a practical, data-driven method to predict wellbore washouts in tight carbonate formations by combining reclassified drilling activity data with high-resolution caliper measurements. The original Drill-State were reorganized into 45 more detailed activity types, and each was given a severity score. Using this, synthetic caliper logs were created and then compared with historical LWD data. Even though the workflow was only tested in a desktop environment using Excel simulation and past data, the comparison showed that the method is feasible.

Although this work has not yet been tested in real-time field operations, it shows strong potential to be included in automated drilling systems. Several challenges were identified, such as time-depth conversion, standardizing Drill-State, and software integration, which need to be solved for full real-time application. Moving forward, the next steps will focus on automating the classification, embedding it into digital drilling platforms, and testing the approach in actual field operations. In the long term, this workflow can support safer, more efficient, and more environmentally friendly drilling practices. To move this methods closer to field readiness, several actions are recommended:

- **Integrate the reclassification into real-time drilling software**
  Add the 35 refined activity types directly into drilling platforms so that the system can process and calculate severity in real time.

- **Run pilot tests in the field**
  Conduct trial runs under real drilling conditions to check if the synthetic caliper model performs well and is reliable.
- **Use a larger and more diverse dataset**
  Include more wells and different types of formations to make the model stronger and more general—not only for tight carbonates.
- **Build a flexible severity scoring engine**
  Develop a backend system (or API) that can automatically assign severity scores based on activity and real-time well conditions.
- **Automate time-to-depth conversion**
  Use software to handle the conversion of Drill-State logs from time to depth more accurately—especially for complex operations like reaming or static conditions.
- **Collaborate with drilling software vendors and operators**
  Work with industry partners to test and roll out the method in live drilling environments, ensuring it's easy to adopt and works well with existing tools.

Summary

This methodology provides a step-by-step workflow that combines drilling data, borehole shape information, and workflow to predict wellbore washouts. The process starts with data collection and synchronization, then continues to feature engineering and modeling, and ends with validation and preparation for deployment.

Different from traditional methods that usually analyze caliper logs only after drilling, this approach allows early detection of washout risk, so the drilling team can take action in real time/ timely after drilling finished. It also gives a strong foundation to be integrated into a digital drilling advisory system. In the future, this method can be improved further by adding more wells into the dataset, using machine learning models, and combining rock mechanical properties from LWD to improve prediction accuracy and make the model more general for other formations.

Abbreviations

LWD: Logging-While-Drilling (technology)
ROP: Rate of Penetration (drilling parameter)
WOB: Weight-on-Bit (drilling parameter)
FDP: Field Development Planning (process)
XML: Extensible Markup Language (format for tagging and structuring data)

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